TOWARDS SCALABLE QUADRUPED IMITATION: LEARNING ANIMAL MOTIONS FROM MONOCULAR VIDEOS WITHOUT MOTION CAPTURE

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ABSTRACT. We present a framework to retarget animal motions from monocular dog videos to a quadruped robot without relying on motion capture or manual annotations. Our approach combines a learned skeleton representation with improved video-to-joint pose estimation to reconstruct animal motion. We then train a locomotion policy using these reconstructed motions as demonstrations, enabling joint-level imitation on a physical robot. Unlike prior methods that require structured motion datasets or predefined keypoints, our framework scales across diverse dog breeds and videos. We demonstrate in our experiments that the resulting policies produce agile and realistic behaviors on a simulated robot, highlighting the potential of our method for scalable, vision-based motion imitation. Our code is available at https://github.com/TSUITUENYUE/motion-imitation. **For links in the paper to be functional, please download the report for viewing.**

1. INTRODUCTION

Mimicking animal locomotion from videos remains a compelling challenge in robotics due to the diversity of morphologies and movement styles across species. Traditional motion retargeting approaches, such as those applied to humanoids, rely on supervised pipelines that extract joint angles frame-by-frame and craft reward functions to encourage joint-level imitation [17]. While effective, these methods require expensive motion capture setups or manually labeled joint positions.

Similar strategies have been extended to quadrupeds, using either motion capture of animals or curated datasets with skeletal annotations [7, 16]. However, reliance on structured data limits scalability between species and environments. To address this, recent work has explored using high-level visual cues as rewards. For example, Chanesane et al. [5] trained a CNN classifier on animal video datasets to detect action categories such as running or jumping, and used the classifier's confidence scores as a reward signal in reinforcement learning. Although this approach avoids keypoint supervision, it does not aim to mimic specific animal motions.

Motivated by this gap, our work investigates whether it is possible to retarget motions from vastly different dog breeds to a quadruped robot using only monocular videos. We use a template model as a prior to estimate the pose from videos and refine the pose with learnable skinning weights. We then propose a novel method for policy training that leverages the learned skeletons. We also innovate several rewards to encourage faster convergence of motions instead of training from scratch.

Contributions. In this report, our main contributions are:

(1) A learned skeleton model for motion re-targetting without mocap data or human labeled keypoints directly from monocular videos, combining SMAL template priors with learned per-instance skinning weights for improved articulation.
(2) A steady transfer of the re-targetting pipeline from Pybullet to Genesis with Go2.
(3) Several innovated rewards that strengthens the robustness of locomotion while mataining key characterstics from original gait patterns.

2. BACKGROUND

Motion imitation from videos involves three key challenges: (1) reconstructing accurate 3D poses from 2D observations, (2) adapting these poses to a target morphology (retargeting), and (3) learning stable policies that reproduce the motion.

Let $\mathcal{V} = \{I_t\}_{t=1}^{T}$ be a monocular video sequence of an animal. Our goal is to learn a policy π_θ that controls a robot with joint configuration $\mathbf{q} \in \mathbb{R}^n$ to mimic the animal's motion. The pipeline consists of:

Pose Estimation. For each frame I_t, we estimate the animal's 3D joint positions $\mathbf{J}_t \in \mathbb{R}^{3\times k}$ and mesh vertices $\mathbf{V}_t \in \mathbb{R}^{3\times m}$ using the SMAL model with learned skinning weights $\mathbf{W}$.

Motion Retargeting. We solve the optimization problem:

$$\mathbf{q}_t^* = \arg\min_{\mathbf{q}_t} \|\phi(\mathbf{J}_t) - \psi(\mathbf{q}_t)\|^2 + \lambda R(\mathbf{q}_t) \tag{1}$$

where ϕ, ψ map between animal and robot joint spaces, and R enforces kinematic constraints.

Policy Learning. The retargeted motions $\{\mathbf{q}_t^*\}$ serve as demonstrations for training π_θ via RL, using the reward function described in Section 4.3.

3. Related Work

3.1. **Articulated Models Reconstruction.** Recovering an animal's underlying skeleton model from images or videos presents an inherently ill-posed challenge. Compared to humans, animals exhibit greater variation in shape, pose, and appearance, making this task significantly more complex. To address this, researchers have proposed parametric models tailored to specific species [2, 3, 13], which can be fitted to monocular images or videos. However, constructing accurate template models remains costly and labor-intensive, particularly for diverse animal classes. As an alternative, model-free approaches learn animal shape representations by deforming simple geometric primitives, such as spheres [4, 8, 23]. Other recent methods construct articulated models directly from videos using animatable neural fields or 3d gaussians[25, 27, 12].

In our work, we adopt the SMAL model [31] as a strong prior and optimize its parameters to fit monocular video data. To improve articulation quality, we additionally learn per-instance skinning weights during the fitting process.

3.2. **Motion Imitation.** Recently, motion imitation using reinforcement learning has emerged as a popular and rapidly advancing field. For humanoid robots, it is relatively straightforward to teach behavior by learning from human motion data. Several works, such as [18, 9, 22, 26, 10, 6], have achieved notable success in this area. However, motion imitation for legged robots remains a challenging problem due to the complexity of their dynamics, limited availability of high-quality animal datasets, high sample complexity, and the structural discrepancy between rigid robots and flexible animal bodies.

Although prior works have made significant progress in addressing these challenges, limitations still persist. For example, Pan et al.[15] circumvent the need for animal motion data by having humans demonstrate legged behaviors, which restricts the diversity and richness of motions available for imitation. SLoMo[30] learns directly from animal videos but relies on manual motion scaling and oversimplified dynamics, which can strip away important animal-specific characteristics. The most closely related work to ours is by Peng et al. [16], which successfully extracts motion data from real-world animal videos for use in legged robot imitation.

In our work, we utilize a simulation platform to accelerate the sampling process inherent in reinforcement learning. Additionally, we introduce a novel keypoint retargeting strategy to bridge the structural gap between animal motions and robot morphology. To better capture the unique behavioral traits of different animals, we also design a specialized reward function that guides the robot to emphasize species-specific motions during imitation.

3.3. **Locomotion Training via Reinforcement Learning and simulations.** Deep Reinfocement Learning has been shown to provide promising locomotion training recently [14], [11], [21]. RL integration in Issac Gym and Issac Labs has been made easy with legged gym [19]. The legged gym allows for a clean implementation of PPO and allows us to shift efforts to retargeting and modifying rewards from it. However, we are utilizing a RTX 5090 GPU which does not yet support Issac Gym due to version compatibility. Therefore, we are utilizing Genesis [1], which is a faster and newer simulation that has an integrated legged gym from [19].

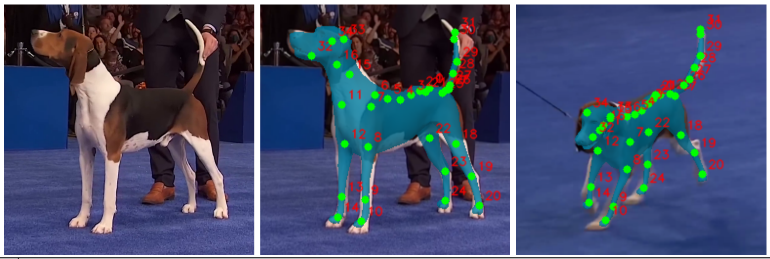
4. Approach

4.1. **Learning Skeleton from Monocular Videos.** Learning accurate animal skeletal structures from monocular videos is inherently challenging due to the high degree of ambiguity and the lack of depth information. To address this, we leverage the SMAL category-specific template model [31] as a strong structural prior. For initial pose estimation, we follow the BARC pipeline [20], which estimates per-frame 3D poses using the SMAL model. This provides a coarse but consistent initialization for downstream processing.

We then fit an articulated 3D mesh to the entire video sequence by optimizing both the pose and the linear blend skinning (LBS) weights in GART[12] baseline. These learned skinning weights improve articulation fidelity and enable temporally consistent mesh deformations across frames. To facilitate motion retargeting onto robotic platforms, we normalize the motion data into a consistent world coordinate system. Specifically, we define the root of the skeleton as the center of the pelvis, and align the root position in the first video frame to the origin of the coordinate system.

We apply the same translation offset to all joints across the sequence, ensuring that all joint positions and rotations are expressed relative to the same world frame with the first-frame root as the origin.

From each frame, we extract the 3D joint positions and their orientations, the latter represented as quaternions. Each joint's state is thus represented using seven floating-point values—three for position and four for rotation. For our application of motion retargeting to the Unitree Go2 quadruped robot [24], we only retain the 9 joints that correspond to the robot's actuation structure. Detailed joints usage are listed in fig. 1:



Name	root	l shoulder	l paw	r shoulder	r paw	l hip	l foot	r hip	r foot
Index	0	8	10	12	14	18	20	22	24

FIGURE 1. Predicted mesh and joint position of a hound breed. Top: Visualizations of predicted mesh and joint positions. Bottom: Joint index correspondence table. The joints highlighted in table were primarily used for motion retargeting.

4.2. **Motion Retargeting.** To adapt motion data recorded from the animal subject to our robot with different morphology, we followed the strategy from [16] to apply inverse kinematics (IK) for motion retargeting to fit robot's morphology. With the cooresponding pairs of keypoints between source model and target, at each timestep, the source motion specifies the 3D position $\hat{\mathbf{x}}_i(t)$ for each keypoint i, and the robot's target keypoint position $\mathbf{x}_i(\mathbf{q}_t)$ is computed as a function of the robot's pose $\mathbf{q}_t$, expressed in generalized coordinates. We then solve an optimization problem to determine the sequence of robot poses $\mathbf{q}_{0:T}$ that minimize the discrepancy between source and target keypoint positions, while also penalizing deviations from a predefined nominal pose $\bar{\mathbf{q}}$. The objective function is formulated as:

$$\arg\min_{\mathbf{q}_{0:T}} \sum_t \sum_i \|\hat{\mathbf{x}}_i(t) - \mathbf{x}_i(\mathbf{q}_t)\|^2 + (\bar{\mathbf{q}} - \mathbf{q}_t)^T \mathbf{W} (\bar{\mathbf{q}} - \mathbf{q}_t), \tag{2}$$

where $\mathbf{W}$ is a diagonal matrix of regularization weights. This formulation ensures accurate tracking of keypoints while maintaining naturalistic and feasible robot motions.

After the retargeting process complete, we are able to get the position and orientation of quadruped robot's root and the joint angles for all 12 movable joints. The video of motion retargeting is listed in table 1.

4.3. **RL-based Imitation Learning.** We treat imitation learning as a reinforcement-learning (RL) problem. At each timestep t the agent observes state s_t, executes action $a_t \sim \pi(a_t \mid s_t)$, transitions to s_{t+1} and receives reward $r_t = r(s_t, a_t, s_{t+1})$. The objective is to maximise

$$J(\pi) = \mathbb{E}_{\tau \sim p(\tau|\pi)} \Big[\sum_{t=0}^{T-1} \gamma^t r_t \Big], \quad p(\tau \mid \pi) = p(s_0) \prod_{t=0}^{T-1} p(s_{t+1} \mid s_t, a_t) \pi(a_t \mid s_t). \tag{3}$$

The reference motion provides desired joint angles $\hat{q}_t$ and velocities $\hat{\dot{q}}_t$. Our final reward is a combination of original rewards used in [7], and several innovative custom rewards to enhance and accelerate performance and encourage faster convergence.

1. Joint-pose matching

$$r_t^{\text{pose}} = \exp\Big[-\tfrac{1}{0.20} \sum_j w_j (q_t^j - \hat{q}_t^j)^2\Big], \tag{4}$$

where w_j up-weight the six front-leg joints (hip, thigh, calf) to emphasise gait generation.

2. Joint-velocity matching

$$r_t^{\text{vel}} = \exp\Big[-\tfrac{1}{0.15} \sum_j w_j^v (\dot{q}_t^j - \hat{\dot{q}}_t^j)^2\Big], \tag{5}$$

with larger weights on hip and thigh joints to capture cyclic rhythm.

3. End-effector (foot) matching After obtaining the joint rotations based on joint coordinates, we ran the inverse kinematics to estimate the location of thigh and calf links for both reference pose and rl agent. Note that in genesis the IK is done on links rather than joints which is why we modified this reward.

$$r_t^{\text{ee}} = \exp\Big[-\tfrac{1}{0.30} \sum_{j\in\text{thigh, calf}} (q_t^j - \hat{q}_t^j)^2\Big]. \tag{6}$$

4. Base stability This reward penalises bouncing and roll/pitch:

$$r_t^{\text{stab}} = \tfrac{1}{3}\Big[e^{-|v_z|/0.03} + e^{-(\omega_x^2+\omega_y^2)/0.1} + e^{-|h-h_{\text{target}}|/0.05}\Big]. \tag{7}$$

5. Absolute-position progress This one rewards sustained forward displacement Δx while increasingly penalising side drift Δy. We incorporate this reward because the robot tends to stand stil of move in place with basic rewards from [7]

$$r_t^{\text{abs}} = 1 - \underbrace{\max(0, 0.3 - \Delta x)}_{\text{lack-of-progress}} - \underbrace{0.4\,|\Delta y|}_{\text{side drift}} - \dots \tag{8}$$

6. Straight-line motion This reward encourages matching a progressively higher speed target v_x^* and suppresses lateral motion and yaw. It encourage the robot to further improve its speed as it gets better at walking.

$$r_t^{\text{str}} = e^{-|v_x - v_x^*|/0.3} - \Big(e^{-|v_y|/0.03} - 1\Big)\,\beta - \Big(e^{-|\omega_z|/0.07} - 1\Big)\,\beta, \tag{9}$$

$$\beta = 1 + 4(v_x^* - 0.5). \tag{10}$$

7. Front-leg activity During preliminary tests, we found that relying on joint velocity will encourage forward motion resulting in the agent to simply leaning forward and fall. These often lead to a slower convergence, as the it was able to gain reward from velocity matching but will take a much longer episode to move forward. To guide the agent, we created a front-leg activity to ensure that the front legs are attempting to push the robot forward.

$$r_t^{\text{front}} = \underbrace{\text{norm}\big(\dot{q}_{\text{front}}\big)}_{\in[0,1]} + 0.5\,\mathbf{1}_{\text{alt. hip signs}} + 0.3\,\mathbf{1}_{\text{any movement}}. \tag{11}$$

Finally, We also included domain randomization to allow sim2real transmission. At every reset, we uniformly sample friction, restitution, gravity and link masses within the ranges $[0.9, 1.1]$, $[0, 0.05]$, $g \in [-9.9, -9.7]$ m/s^2 and $[0.95, 1.05]$ respectively.

5. Experimental Results

5.1. **Skeleton Mapping.** We tested our pipeline on short, publicly-available dog videos scraped from the web. Despite the unknown camera parameters and wide variation in breed and size, the model still reconstructs a coherent mesh and extracts a kinematically-consistent skeleton suitable for motion-capture–style retargeting. The key challenge arises when mapping these joints to Go2 commands, as each animal requires individual offset and scale calibration. While this manual process shows promise, it currently remains undone given the limited time scope of this project as it is inherently labor-intensive. Empirically we found that body length, limb proportions and even species (dog vs. wolf) all shift the hip and shoulder reference frames, so a light calibration pass is still necessary before deployment. **Examples:** Shiba Inu, Pit bull, German Shepherd

TABLE 1. Video Recordings of Retargeting Comparisons

Platform	Trot	Trot-2	Pace	Canter
PyBullet	link	link	link	link
Genesis	link	link	link	link
RL Policy	link	link	link	link

5.2. **Retargeting.** Due to the tricky scale-offset mapping between the learned skeleton model and the retargeting agent mentioned above, our current experiment is mostly based on the mocap data from a dog [16]. There are 6 defined motions: pace, trot, trot2, tanter, left turn, and right turn. The link to the motion retarget video is embedded here. The go2 agent is able to imitate the dog's movement pretty well. We also transformed the retargeted pose in Genesis which later used for our PPO training as demonstrated in fig. 2.

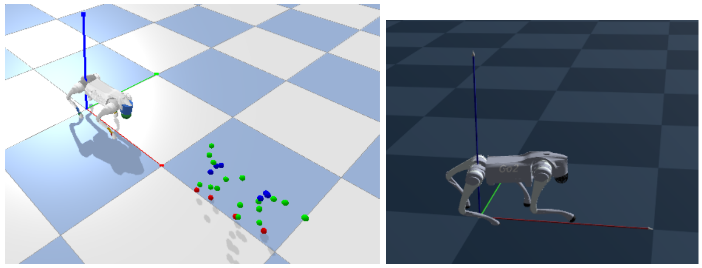


FIGURE 2. A demonstration on retargeting pose in pybullet and genesis.

5.3. **RL trained policy.** One of the key difficulties in RL training is modifying the weights of different rewards. Most of our time is spent tuning these weights, and emphasizing the absolute position reward is key to encouraging forward motion. After training 500 iterations per mocap sequence, we observed that the resulting policies preserved key characteristics of the original motion while improving physical plausibility. For example, in fig. 4, the Canter sequence originally involved the left front leg hovering above the ground throughout the motion; in the RL-trained version, the leg still lifts high, but now contributes to propulsion by bringing it back down. Similarly, the Pace policy retains a slower stepping rhythm than Trot2, reflecting distinct gait signatures. A typical training is approximately 20 minutes on a RTX 4070 GPU, however fig. 3 suggests that we could reduce the iterations numbers as the first four key rewards plateau quickly. The recorded footage from the retageting and RL trained based on 4 mocap-based motions are shown in table 1.

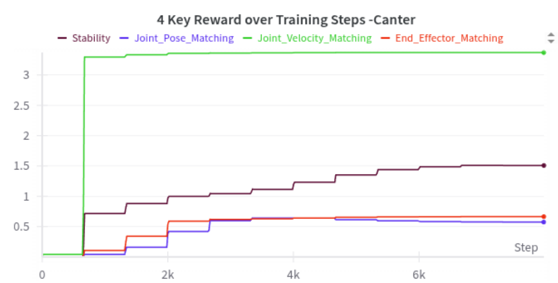


FIGURE 3. Reward vs training time steps

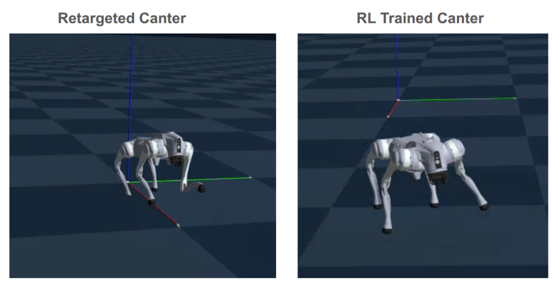


FIGURE 4. Left Leg Lift maintained from retarget to RL

6. DISCUSSION

6.1. **Skeleton Prior.** Our pipeline still depends on SMAL's[31] hand-crafted kinematic prior through BARC[20]. This becomes limiting when templates are unavailable or animals deviate from the model. Two approaches might relax this constraint:

(1) **Prior-free skeleton discovery**. Recent work[28, 29] shows skeletons can be inferred from motion data alone. Extending this to quadrupeds would eliminate reliance on SMAL's template.
(2) **Instance-specific refinement**. Optimizing per-sequence bone lengths and joint offsets, with light anatomical constraints, could better handle atypical proportions.

Currently, the rigid prior sometimes causes implausible poses. Making it adaptive would improve accuracy for non-standard subjects and robustness in difficult conditions.

6.2. **Physcially Accurate Retargetting.** Although RL policies preserve characteristic features of the re-targetted motion, they do not perfectly replicate the original trajectories. This is expected as the re-targetted motions are not physically grounded—they merely replay kinematic sequences captured from mocap. In contrast, RL policies must operate within the physical limits of the simulated dog body and environment. Consequently, deviations often reflect physically plausible adaptations rather than failure and highlight the importance of optimizing for function, not just form.

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